A COMPOSITE OF LOCALLY QUASI-FINITE MORPHISMS OF ALGEBRAIC STACKS IS LOCALLY QUASI-FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The base change permanence of local quasi-finiteness. Invoked at the start of the counted proof (source lines 5045-5048). Makes the two morphisms after base change to a field locally quasi-finite.
- The characterization of quasi-DM stacks through flat locally quasi-finite covers by schemes. Invoked inside the counted proof (source lines 5049-5056). Produces the scheme covering the middle stack.
- The statement that a scheme locally quasi-finite over a field has discrete space. Invoked inside the counted proof (source lines 5057-5060). Gives discreteness of the presenting scheme.
- The structure of a scheme locally quasi-finite over a field as a disjoint union of spectra of Artinian local algebras. Invoked near the end of the counted proof (source lines 5064-5074) as a cross-chapter pointer. Reduces discreteness of the fibre to the residue field points of the pieces.

Source reference. Original lemma lemma-base-change-locally-quasi-finite: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4989–4989. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-locally-quasi-finite, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-composition-finite-type: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3488–3488. Navigation pointer only: the counted text refers to the Stacks item lemma-composition-finite-type, which is an existing labeled result of this same chapter and is not part of

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Source reference. Original lemma lemma-composition-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 852–852. Navigation pointer only: the counted text refers to the Stacks item lemma-composition-separated, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-locally-quasi-finite-over-field: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5002–5002. Navigation pointer only: the counted text refers to the Stacks item lemma-locally-quasi-finite-over-field, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-separated-implies-morphism-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1061–1061. Navigation pointer only: the counted text refers to the Stacks item lemma-separated-implies-morphism-separated, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original theorem theorem-quasi-DM: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4286–4286. Navigation pointer only: the counted text refers to the Stacks item theorem-quasi-DM, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference varieties-lemma-algebraic-scheme-dim-0: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4440–4440. Navigation pointer only: the counted text refers to the Stacks item varieties-lemma-algebraic-scheme-dim-0, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *A composition of a locally quasi-finite morphisms is locally quasi-finite.*

Proof. We have seen this for quasi-DM morphisms in Lemma lemma-composition-separated and for locally finite type morphisms in Lemma lemma-composition-finite-type. Let $\mathcal{X} \to \mathcal{Y}$ and $\mathcal{Y} \to \mathcal{Z}$ be locally quasi-finite. Let k be a field and let $\mathrm{Spec}(k) \to \mathcal{Z}$ be a morphism. It suffices to show that $|\mathcal{X}_k|$ is discrete. By Lemma lemma-base-change-locally-quasi-finite the morphisms $\mathcal{X}_k \to \mathcal{Y}_k$ and $\mathcal{Y}_k \to \mathrm{Spec}(k)$ are locally quasi-finite. In particular we see that $\mathcal{Y}_k$ is a quasi-DM algebraic stack, see Lemma lemma-separated-implies-morphism-separated. By Theorem theorem-quasi-DM we can find a scheme V and a surjective, flat, locally finitely presented, locally quasi-finite morphism $V \to \mathcal{Y}_k$. By Lemma lemma-locally-quasi-finite-over-field we see that V is locally quasi-finite over k, in particular $|V|$ is discrete. The morphism $V \times_{\mathcal{Y}_k} \mathcal{X}_k \to \mathcal{X}_k$ is surjective, flat, and locally of finite presentation hence $|V \times_{\mathcal{Y}_k} \mathcal{X}_k| \to |\mathcal{X}_k|$ is surjective and open. Thus it suffices to show that $|V \times_{\mathcal{Y}_k} \mathcal{X}_k|$ is discrete. Note that V is a disjoint union of spectra of Artinian local k-algebras

A_i with residue fields k_i, see Varieties, Lemma varieties-lemma-algebraic-scheme-dim-0. Thus it suffices to show that each

$$|\operatorname{Spec}(A_i) \times_{\mathcal{Y}_k} \mathcal{X}_k| = |\operatorname{Spec}(k_i) \times_{\mathcal{Y}_k} \mathcal{X}_k| = |\operatorname{Spec}(k_i) \times_{\mathcal{Y}} \mathcal{X}|$$

is discrete, which follows from the assumption that $\mathcal{X} \to \mathcal{Y}$ is locally quasi-finite. $\square$